

Kostas Powell

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EDUCATION

QUEEN'S UNIVERSITY

Bachelors of Applied Science (BASc), Mechanical Engineering

Certificate in Data Analytics (School of Computing)

Expected Graduation: 2028

Cumulative GPA: 3.6/4.3 | Dean's List Recipient

Kingston, ON

September 2024 - Present

TECHNICAL PROJECTS

HYBRID ROCKET INJECTOR DESIGN - QUEEN'S ROCKET ENGINEERING TEAM

- Designed, simulated, and validated a multi-orifice swirl injector for a hybrid rocket engine.
- Built a Python-based thermodynamic and two-phase flow analysis tool using CoolProp to analyze test and simulation data.
- Performed CFD and FEA in ANSYS to validate flow behavior and structural integrity under operating loads.
- Produced GD&T-compliant technical drawings and CAD within SolidWorks for 5-axis CNC manufacturing.
- Collaborated with cross-functional team members to troubleshoot performance issues and implement improvements.

HIGH ALTITUDE WEATHER BALLOON POWER SYSTEM

- Designed and assembled a battery management and power distribution system for extreme cold environments.
- Integrated sensors, power components, heating elements to ensure reliable system operation.
- Programmed an arduino controlled feedback loop for regulating battery temperature.
- Performed thermal analysis and validation testing, documenting results for design iteration.

ENGINE PERFORMANCE AND 6-DOF FLIGHT SIMULATOR

- Built a MATLAB aircraft engine performance simulation to analyze thrust and specific impulse across operating conditions.
- Integrated propulsion model into a 6-DOF flight dynamics simulation.
- Analyzed performance trends using structured datasets and parametric sweeps.
- Verified model behavior through comparison of analytical, numerical, and simulated results.

STIRLING ENGINE FABRICATION

- Designed and fabricated a functional Stirling engine using manual machining and welding.
- Performed tolerance verification, and functional testing to validate mechanical performance.

TECHNICAL SKILLS

CAD: SolidWorks, Fusion360.

Simulation and Analysis: ANSYS (CFD, FEA), MATLAB.

Programming: Python (NumPy, SciPy, CoolProp), C, MATLAB.

Manufacturing: Manual machining, welding, CNC machining, GD&T, technical drawings.

Electronics: Circuit design, Arduino, soldering.

Data & Documentation: Excel (data analysis, plots), technical reports, test documentation.

WORK EXPERIENCE

CYCLING AND TRIATHLON COACH - EXCELERATION, TRIATHLON BC

- Developed structured training plans based on performance data and athlete feedback.
- Communicated technical information clearly to athletes and support staff

Vancouver, BC

September 2022 - August 2025

ACTIVITIES AND ATHLETICS

- Queen's Rocket Engineering Team - Motor Subteam Lead.
- Queen's Varsity Triathlon Team - Athlete and VP of Logistics.
- Nationally Ranked High-Performance Triathlete - Junior Nationals Qualifier (2024).
- Queen's Engineering Orientation Week - Head of OTIS.